

Towards Reliable Solar Plant Generation Forecasting: Temporal Context Self-Attention Across Recurrent and Transformer Models

Supplementary Material for Review — code, curated artifacts, and interactive dashboard supporting the manuscript submitted to *IEEE Transactions on Industrial Informatics*.

This repository contains the complete research pipeline behind the **Temporal Context Self-Attention (TCSA)** mechanism proposed in the manuscript, together with the exported figures, metric tables, trained-model checkpoints, and the web-based **Interactive Analytical Dashboard (IAD)** described in Section IV-D. It is provided so that reviewers can trace every reported number and figure back to the code that produced it.

► Interactive Analytical Dashboard (live)

<https://xundullah.github.io/Demo--TCSA-SolarGen-Prediction/>

This is the **IAD of Fig. 6** in the manuscript, and the single most direct way to inspect the results without running any code. The dashboard — *SPFS: Solar-Plant Forecasting Simulator* — replays the day-ahead predictions of the baseline, MHSA, and TCSA variants against the measured generation, hour by hour, over the 144-hour test horizon, for each of the three solar plants and each architecture group.

The dashboard provides:

- **Simulation setup** — selection of the Solar Plant (SP) and the architecture family (LSTM, GRU, Transformer), with play, step, next-peak, and speed controls.
- **System log** — operational events, including sunrise infeed onset, sunset cessation, and the daily generation peak with the corresponding TCSA deviation.
- **Data log** — the current-hour measured output alongside the baseline, MHSA, and TCSA predictions.
- **Prediction-versus-actual chart** — hour-by-hour traces, with not-yet-played hours dimmed.
- **Live cumulative error** — MAE, RMSE, and R^2 of the proposed model updated over the hours played so far, so the error can be observed as it evolves through the day rather than as a single aggregate value.
- **Single-line diagram** — the multi-plant electrical architecture of Fig. 1, in which each component reports its parameters interactively.

No installation is required. The same page is `index.html` in this repository and runs locally by opening that file directly in a browser — no build step, no server.

1. Scope and correspondence with the manuscript

The manuscript is organized around three domains; this repository follows the same organization.

Manuscript domain	Manuscript modules	Where it lives in this repository
DE — Data Engineering (Sec. III-A, Algorithm 1)	RHD, HDF, DCI	Code/Site02_DataPreparation.ipynb, Library/dataProcessing.py, Library/dataAnalysis.py
FMA — Forecasting Model Architecture (Sec. III-B, Algorithms 2–3)	RSN family, TEN family	Code/Site02_ModelDevelopment.ipynb, Library/modelDevelopment.py
FEV — Forecast Evaluation & Visualization (Sec. IV)	MPV, DHF, IAD	Code/Site02_ResultAnalysis.ipynb, Library/modelEvaluation.py, index.html

Naming map

Internal identifiers in the code predate the manuscript notation. The correspondence is:

In this repository	In the manuscript
Site-02, Site#02	The site in Gyeongju-si, Gyeongsangbuk-do, South Korea (Sec. II)
Plant ...G0001	Solar Plant I (SP 1)
Plant ...G0003	Solar Plant II (SP 2)
Plant ...G0004	Solar Plant III (SP 3)
Station ...G4001	The co-located weather station (Sec. II)
PkW	Plant output P_{pv}^{SP} in kW

Base / MHSA / TCSA	Variant $\zeta \in \{\text{Base, MHSA, TCSA}\}$ of (4) and (5)
skill score	Prediction Skill (PS) of (6)

2. The eight benchmarked models

The TCSA mechanism is applied to two model families — Recurrent Sequence Networks (RSN) and Transformer Encoder Networks (TEN) — and compared against the baseline and conventional multi-head self-attention (MHSA) counterparts under a common experimental setting. All models share the Day-Ahead Hourly Forecasting (DHF) task: a 168-hour (one-week) input window of plant output plus six meteorological variables, mapped to the next 24 hourly outputs.

Family	Cell / encoder	Baseline	MHSA (typical)	TCSA (proposed)
RSN	LSTM	✓	✓	✓
RSN	GRU	✓	✓	✓
TEN	Transformer encoder	—	✓ (family reference)	✓

The Transformer group has no plain baseline, since its encoder is already attention-based; the MHSA variant therefore serves as its family reference for the Prediction Skill (PS) computation. Every model is trained and validated independently for each of the three plants, giving 24 SP-model pairings.

3. Reproduction pipeline

Run the notebooks in `Code/` in this order.

3.1 Site02_DataPreparation.ipynb — the DE layer

Implements Algorithm 1. Loads the pre-processed hourly record from `Export/Data/Site#02_Data.zip`, removes observations outside the physical bounds, partitions the record chronologically, reconstructs the remaining gaps, applies the variable-wise Min-Max scaler fit on the training partition alone, and windows the result per plant.

Gap reconstruction follows (1): a harmonic-climatology fit with shifted-residual transfer for the meteorological variables, and diurnal bootstrapping — uniform sampling from the same-month, same-hour training pool — for the PV power, so that night-time zeros and the sunrise/sunset ramp are preserved by construction. Implemented in `Library/dataProcessing.py`.

Also produces the diurnal-pattern figure of the site via `dataAnalysis.plot_diurnal_patterns`.

3.2 Site02_ModelDevelopment.ipynb — the FMA layer

Trains all eight models for each of the three plants under the MSE objective, using the 168 h $\rightarrow$ 24 h windowing scheme of `modelDevelopment.make_windows`. The final 20% of windowed samples form the validation set, separated from the training samples by a 192-hour gap ($\tau_w + \tau_o$) so that no validation window overlaps a training window. Early stopping uses a patience of 20 epochs with best-validation checkpointing. Training histories are written to `Export/History/Site-02/` and the best checkpoints to `Export/Model/Site-02/`.

3.3 Site02_ResultAnalysis.ipynb — the FEV layer

Evaluates every SP-model pairing on the held-out test partition, inverts the normalization so that all reported errors refer to actual plant output in kW, and generates the manuscript figures and metric tables through `Library/modelEvaluation.py`.

4. Figure and table correspondence

All figures are exported as vector PDF and 600 dpi PNG under `Export/Figure/Site-02/`, using a colorblind-safe (Okabe-Ito) IEEE style.

Manuscript	Artifact	Generated by
Fig. 2 — diurnal characteristics of the site dataset	<code>diurnal_patterns.png / .pdf</code>	<code>dataAnalysis.plot_diurnal_patterns</code>
Fig. 3 — training and validation learning curves, 3 plants $\times$ 8 models	<code>LearningCurves-Grid-AllModels-AllPlants.png / .pdf</code>	<code>modelEvaluation.plot_learning_curves_grid</code>
Fig. 4 — test-set metric summary with TCSA improvement	<code>MetricSummary-PkW-AllPlants.png / .pdf</code>	<code>modelEvaluation.plot_metric_summary</code>
Fig. 5 — hourly day-ahead forecast traces versus measured generation	<code>PredictionComparison-PkW-AllPlants.png / .pdf</code>	<code>modelEvaluation.plot_prediction_comparison</code>
Fig. 6 — Interactive Analytical Dashboard	<code>index.html</code> (live demo above)	—

Table II — test-set performance of all models across the three plants	Export/Data/Site-02/PredictionComparison-PkW-AllPlants-Metrics.csv	modelEvaluation.plot_prediction_comparison
TCSA improvement percentages quoted in Sec. IV-B	Export/Data/Site-02/MetricSummary-PkW-AllPlants-TCSA-Improvement.csv	modelEvaluation.plot_metric_summary

Two points a reviewer may wish to verify directly:

1. **All scalar metrics in Fig. 5 are computed over the full test set**, not over the five days displayed in the panels. The displayed window (June 9–13) is illustrative; the per-panel RMSE, MAE, and PS boxes come from the same CSV that backs Table II.
2. **Fig. 3 is the overfitting check**. Each panel annotates the best epoch, the early-stop epoch, the patience window, and the train-validation gap at the best epoch, supporting the claim in Sec. IV-B that all 24 pairings converge without overfitting.

5. Headline results

Reproduced from Table II of the manuscript for orientation; the authoritative values are the exported CSVs.

Plant	Best model	MAE (kW)	RMSE (kW)	rRMSE (%)	R ² (%)	PS (%)
Solar Plant I	TCSA-LSTM	4.74	8.43	42.58	88.63	15.95
Solar Plant II	TCSA-LSTM	4.96	8.74	42.85	87.91	20.15
Solar Plant III	TCSA-LSTM	5.32	9.22	42.07	88.09	15.50

Across all 24 SP-model pairings the TCSA variant is the best of its group without exception: MAE is reduced by up to 28.6% and RMSE by up to 20.2%, while R² increases from 81.03% to 87.91% in the best case. Inference cost is unaffected — every model produces a 24-hour forecast in at most 1.7 ms per window.

6. Repository layout

Code/ Site02_DataPreparation.ipynb Site02_ModelDevelopment.ipynb Site02_ResultAnalysis.ipynb	Research pipeline notebooks DE layer — Algorithm 1: curation, gap reconstruction, scaling, windowing FMA layer — trains 8 models x 3 plants FEV layer — evaluation and manuscript figures
Library/ dataProcessing.py dataAnalysis.py modelDevelopment.py modelEvaluation.py	Reusable modules shared by the notebooks Harmonic-climatology fill for the meteorological variables, diurnal-bootstrap fill for PV power Diurnal-pattern figures, out-of-range removal 168h -> 24h windowing, train/validation split, per-model evaluation, loss-curve plots Full evaluation-figure suite and metrics export
Export/ Data/Site#02_Data.gzip Data/Site-02/ Figure/Site-02/ History/Site-02/ Model/Site-02/	Generated artifacts Pre-processed hourly site dataset (pipeline input) Exported metrics and prediction CSVs All figures (vector PDF + 600 dpi PNG) Per-model training-history pickles Keras checkpoints (best epoch only)
Sim/Site-02/ index.html	Prediction and metric CSVs consumed by the IAD IAD / SPFS - Solar-Plant Forecasting Simulator

7. Environment

The pipeline is built on a TensorFlow/Keras stack. Two interpreters are used because the modelling notebooks are pinned to a GPU-capable TensorFlow release.

Component	Python	Key packages
Site02_DataPreparation.ipynb	3.11	pandas, numpy, matplotlib, scikit-learn
Site02_ModelDevelopment.ipynb, Site02_ResultAnalysis.ipynb	3.9.13	tensorflow-gpu<2.10, tensorflow-addons==0.19.0, pandas, numpy, matplotlib, scikit-learn

The dashboard is dependency-free and runs in any modern browser.

8. Data availability

The raw operational archives are retrieved from the plant operator's SQL database through the connection descriptor of Algorithm 1 and are not redistributed here. The pre-processed hourly record that feeds the published pipeline is included as `Export/Data/Site#02_Data.gzip`, so the DE, FMA, and FEV layers are reproducible end to end from this repository alone.